EXPERIMENT 6: CAPTURE PACKETS FOR ONE MINUTE AND REPORT

Task:T12

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1. Aim

To capture network packets for one minute using Wireshark, identify the top three protocols in the captured traffic, and find the IP address with the highest number of outgoing packets,then write a short report.

2. Course Outcome and Module

Course outcome 3:Analyze live network traffic using Wireshark and identify suspicious activities such as Trojans, backdoors, and sniffers.

Module: Network Traffic Capture and Analysis

3. Tools and Versions

Tool	Version	Operating System
Wireshark	4.6.7	Windows 10 (22H2), 64-bit

Wireshark was used to capture and analyse live network traffic. The Protocol Hierarchy and Endpoints features were used to examine the protocols and IP addresses.

4. Program or Commands

This experiment was performed using the Wireshark (GUI) graphical interface. No separate program or command was required.

The following options were used:

Wi-Fi interface: To capture live network packets.

Statistics → Protocol Hierarchy: To identify the protocols and their packet counts.

Statistics → Endpoints → IPv4: To view the IPv4 addresses and their packet statistics.

Tx Packets: To find the IP address with the highest number of transmitted packets.

5. How to Run

Open Wireshark on the computer.

Select the active Wi-Fi network interface.

Start capturing the live network traffic.

Allow the capture to run for approximately one minute.

Stop the capture.

Save the capture file as SNS_Experiment6_12.pcapng.

Open Statistics → Protocol Hierarchy.

Examine the protocols and their packet counts.

Take a screenshot of the Protocol Hierarchy window.

Open Statistics → Endpoints and select the IPv4 tab.

Click the Tx Packets column heading to sort the transmitted packet counts.

Identify the IP address with the highest Tx Packets value.

Take a screenshot of the IPv4 Endpoints window.

Record the observations and prepare the report.

6. Sample Output with Screenshots

The packets captured during the experiment were analysed using Wireshark. The capture contained 74 packets, and Wireshark reported zero dropped packets.

6.1 Identification of the Top Three Protocols

The Protocol Hierarchy feature was used to examine the protocols present in the captured network traffic. Based on the packet counts shown in the screenshot, the three protocols with the highest counts among the listed protocols, excluding the IPv4 and IPv6 network-layer entries, were TCP, IGMP, and UDP.

Sl. No.	Protocol	Number of Packets
1	Transmission Control Protocol (TCP)	26
2	Internet Group Management Protocol (IGMP)	24
3	User Datagram Protocol (UDP)	12

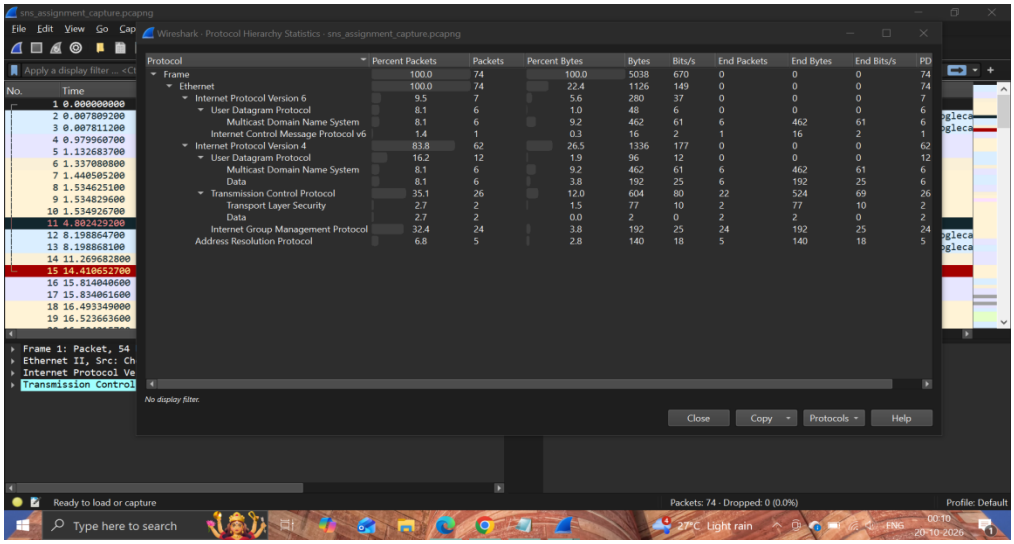


Figure T12-1: Protocol Hierarchy Statistics in Wireshark

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6.2 Identification of the IP Address with the Most Outgoing Packets

The IPv4 Endpoints feature was used to find the IP address that transmitted the highest number of packets. The Tx Packets column was sorted to identify the address with the highest value.

The address 192.168.1.7 had the highest transmitted packet count, with 32 packets.

Parameter	Observed Result
Total packets captured	74
Packets dropped	0
Number of IPv4 endpoints	19
IP address with the highest Tx Packets	192.168.1.7
Transmitted packets from this address	32
Total packets associated with this address	46

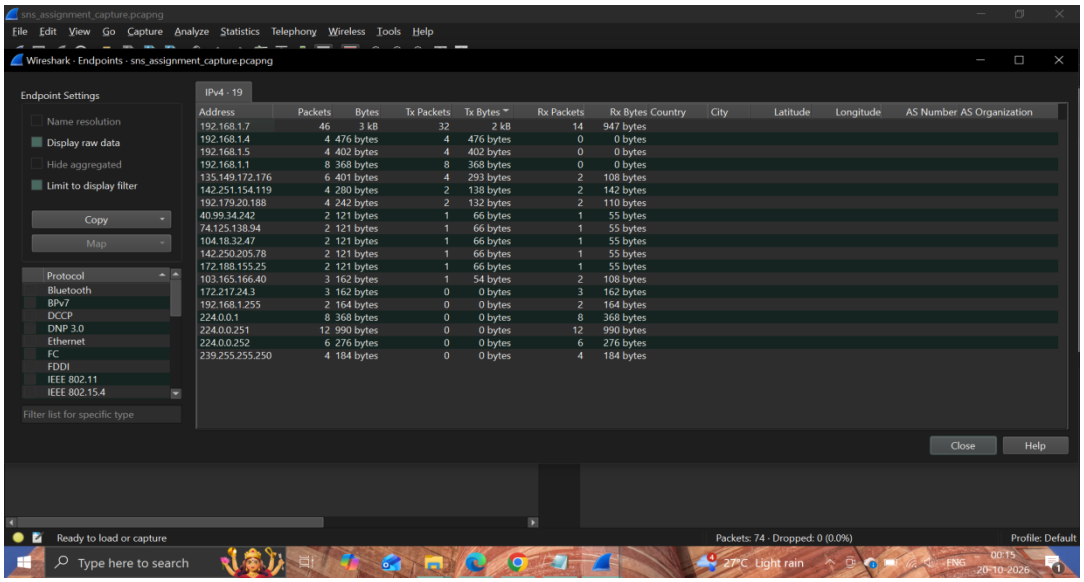


Figure T12-2: IPv4 Endpoints Sorted by Transmitted Packets

6.3 Wireshark Version Information

The About Wireshark window showed that Wireshark version 4.6.7 was installed on Windows 10 (22H2), 64-bit.

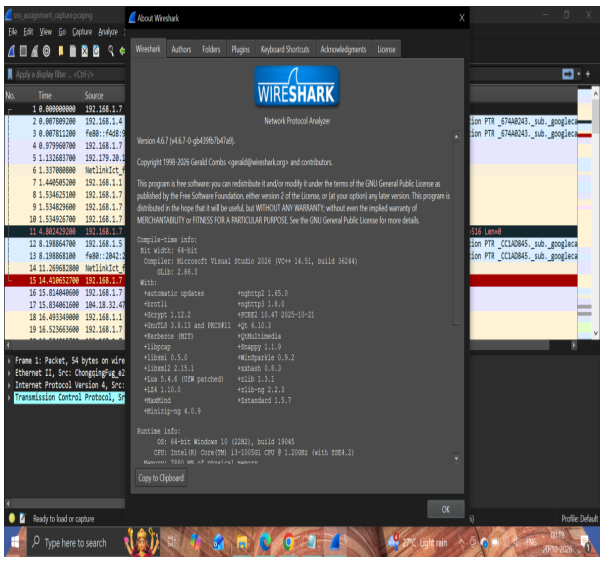


Figure T12-3: Wireshark Version Information

7. Result

Network traffic was captured successfully using Wireshark for approximately 65 seconds. A total of 74 packets were captured, with zero packets dropped.

The three protocols identified by packet count were TCP, IGMP, and UDP, with 26, 24, and 12 packets, respectively. The IPv4 Endpoints analysis showed that 192.168.1.7 had the highest number of transmitted packets, with 32 packets.

Thus, the captured traffic was analysed, the required protocols were identified, and the IP address with the highest outgoing packet count was found.

8. Comparison with Other Versions

The experiment was performed using Wireshark 4.6.7. The following table provides a general comparison between older Wireshark releases and the version used in this experiment. Other versions were not tested practically.

Feature	Older Wireshark Releases	Wireshark 4.6.7
Packet capture	Supports packet capture	Supports packet capture
Protocol analysis	Provides protocol analysis	Provides protocol analysis
Packet statistics	Provides packet statistics	Provides Protocol Hierarchy and Endpoints statistics
User interface	Depends on the release	Interface used in this experiment
Used in this experiment	No	Yes

9. Viva Questions and Answers

1. What is Wireshark?

Wireshark is a network protocol analyser used to capture and examine network packets.

2. What is a network packet?

A network packet is a unit of data transmitted from one device to another through a network.

3. What is Protocol Hierarchy in Wireshark?

Protocol Hierarchy displays the protocols found in the captured traffic along with their packet counts and other statistics.

4. What does Tx Packets mean?

Tx Packets indicates the number of packets transmitted by an endpoint during the capture.

5. What is the difference between TCP and UDP?

TCP is connection-oriented and provides reliable data delivery. UDP is connectionless and does not guarantee that packets will reach their destination.

10. Common Errors: Message, Cause, and Fix

Error or Problem	Possible Cause	Solution
No packets appear	Incorrect network interface selected	Select the active Wi-Fi or Ethernet interface.
Unable to start capture	Permission or configuration issue	Check the capture permissions and network interface settings.

Error or Problem	Possible Cause	Solution
No IPv4 addresses appear	No IPv4 traffic was captured	Check the selected interface and captured packets.
Capture file is not saved	Incorrect save location or filename	Use File → Save As and verify the filename and location.
Tx Packets are not sorted correctly	The column is sorted in ascending order	Click the Tx Packets heading again to change the sorting order.

Declaration: AI tools were utilized for grammatical review, comparison with alternative versions, and assistance in preparing, reviewing, and verifying viva answers.